

More Thermal Energy & Efficiencies

Physics and Mathematics of Sustainable Energy

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1. A home electric generator works by converting the the thermal energy obtained by burning gasoline (petrol) into electrical energy. The Westinghouse WGen7500 has a fuel capacity of 6.6 gallons of gasoline. The generator's efficiency is 18%.
 - (a) How much thermal energy is produced from burning 6.6 gallons of gasoline (petrol)?
 - (b) What are the emissions associated with this?
 - (c) How much electrical energy can the generator generate from 6.6 gallons of gasoline?
 - (d) What are the emissions (in grams per kWh) associated with this electricity?
 2. You use a propane heater to heat the first floor of your house. The efficiency of this heater is 0.8. On a cold winter day you burn 5.5 gallons of propane
 - (a) How much thermal energy is released when you burn this amount of propane?
 - (b) How much of this thermal energy ends up in you home? (Note: this is called the *Heating Load* of the home.
 - (c) Suppose you decide to buy a new heater with an efficiency of 0.9. How much fuel would you need to burn with this more efficient heater to get the same amount of heat in your house?
- $1 \text{ kWh} = 3.6 \text{ MJ} = 3412 \text{ BTU}$
 - $1 \text{ MMBTU} = 1,000,000 \text{ BTU}$
 - Calorific value of heating oil: 12.8 kWh/kg , 37.3 MJ/L , $139,000 \text{ BTU/gallon}$
 - Carbon intensity of heating oil: $260 \text{ g of CO}_2 \text{ per kWh of thermal energy}$. $10.2 \text{ kg CO}_2 \text{ per gallon of fuel}$. $74.1 \text{ kg CO}_2 \text{ per MMBTU of thermal energy}$.
 - Calorific value of petrol: 33 kWh/gal .
 - Carbon intensity of petrol: approx 10.5 kg/gal .
 - $1 \text{ gallon} = 3.8 \text{ liters}$
 - Calorific value of propane: $90,000 \text{ BTU/gal}$.
 - Current average cost of heating oil in Maine: $\$6.00/\text{gallon}$.